Final Project

Report_synthesis

```
============================================================

      Instance                      Module              Cell Count   Cell Area  Net Area   Total Area
-----------------------------------------------------------------------------------------------------
cpu_32bit_top                                                19608 1097735.648 22505.866  1120241.514
  U_CORE                cpu_32bit                          19525   67731.886 22147.415    89879.301
    DP                  cpu_32bit__dp                      11821   36244.391 13341.346    49585.737
      ACC_reg           reg_w_enable_WIDTH32_223              36     330.723     1.482      332.205
      ALU               cpu_32bit__alu                     11588   35006.769 13090.678    48097.447
        add_74_37       add_unsigned                         222     424.678   202.565      627.243
        div_128_37      divide_unsigned_rtlopto_model_9     9456   29285.956 10261.919    39547.875
        mul_116_37      mult_unsigned                        969    3312.432  1293.730     4606.162
        sll_94_28       shift_left_vlog_unsigned             158     338.272   179.339      517.612
        srl_104_28      shift_right_vlog_unsigned            161     336.519   180.841      517.360
        sub_84_37       sub_unsigned                         208     372.276   182.076      554.352
      IR_reg            reg_w_enable_WIDTH32_224              13      95.427     0.000       95.427
      MAR_reg           reg_w_enable_WIDTH7                    9      74.223     0.000       74.223
      MDR_reg           reg_w_enable_WIDTH32                  34     315.333     0.000      315.333
      OVERFL_reg        reg_w_enable_WIDTH1_222                2      10.269     0.000       10.269
      PC_reg            reg_w_enable_WIDTH7_225                9      62.253     0.000       62.253
      ZERO_reg          reg_w_enable_WIDTH1                    2      10.269     0.000       10.269
    FSM                 cpu_32bit__fsm                        63     135.141    62.016      197.157
    ram_wrapper         CDK_S128x16_wrapper                 7641   31352.354  8661.415    40013.769
      u_ram128x16_soft  CDK_S128x16_SYNTHESIZE_RAM          7641   31352.354  8661.415    40013.769
  U_IO                  io_wrapper                            83 1030003.762     0.741  1030004.503
```

Top design: cpu_32bit_top

U_CORE

- DP :11942
- Ram_wrapper: 7487
- FSM: 65

U_IO : io_wrapper : 83 (78 IO_pad , 4 corner cell, 1 CTS BUFF8x)

- How the floorplan size was chosen?

➔first die will be determined number ofports. We have 82 ports + 1 CTS root. So top and bottom will be placed 20 ports for top and bottom and 19 ports for left and right.

➔The design is IO Limited

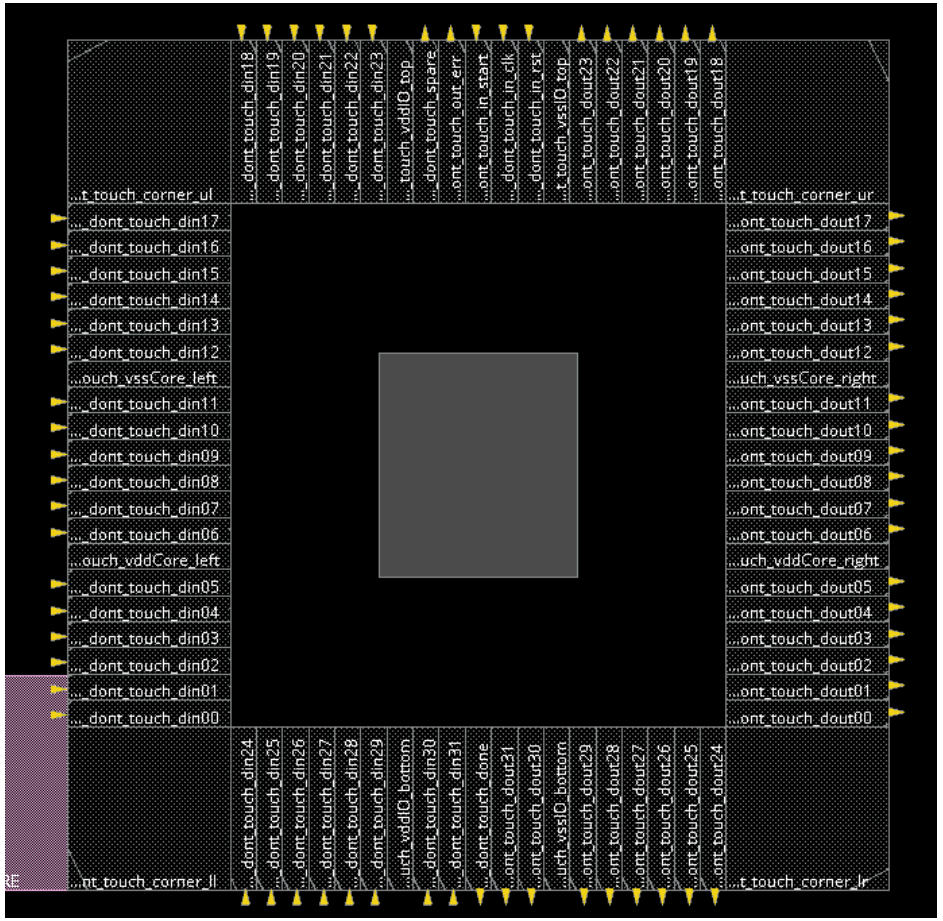
➔So Width die = sum 20*dy ports + 2 corner ports = 1300

➔Length die = 250*2 + 19*dy port = 1260

➔Core size = standard cell / utilization

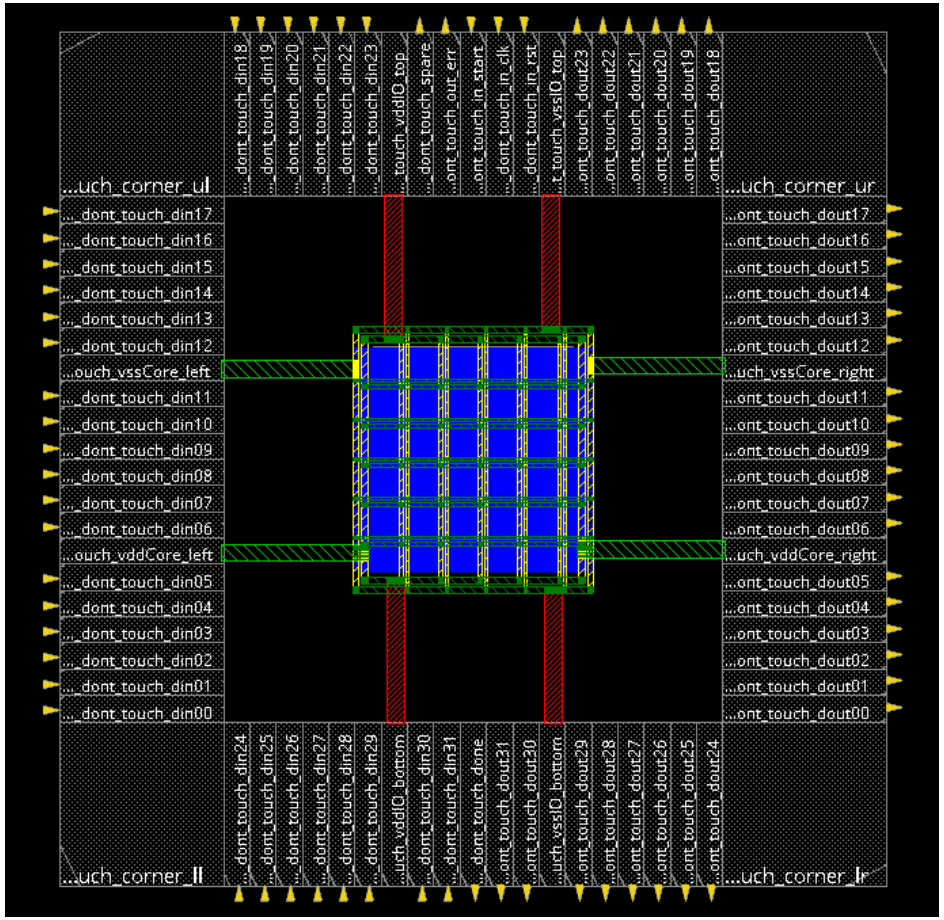
➔I choose utilization with 65%

Floorplan

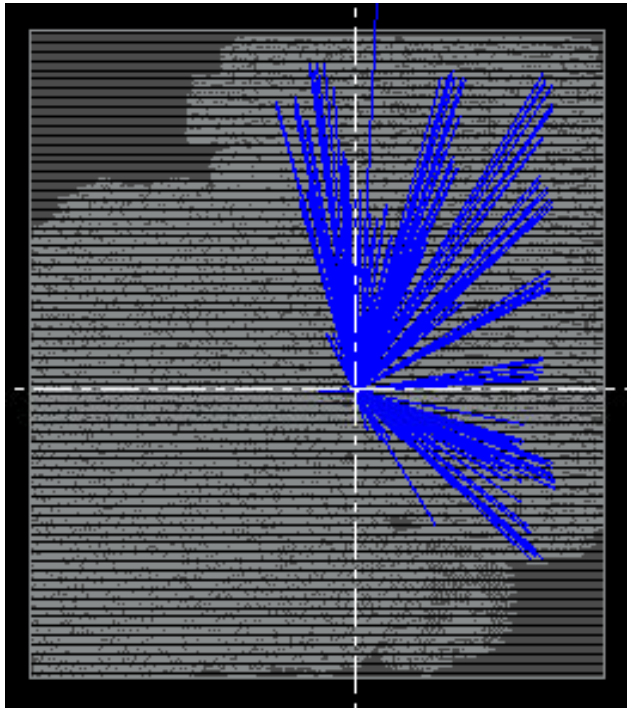


How IO placement was decided?
-> So I will choose 20 ports and 19 ports as above. Din ports will placed left and top left . Dout ports will placed right and top right.

Power

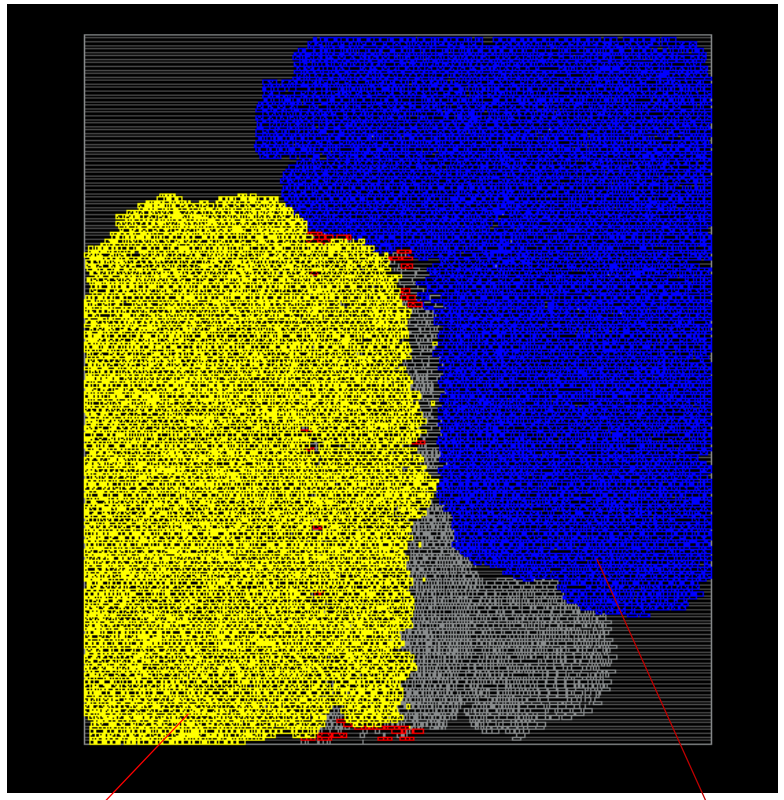

...uch_corner_ul
...dont_touch_din18
...dont_touch_din19
...dont_touch_din20
...dont_touch_din21
...dont_touch_din22
...dont_touch_din23
...touch_vddIO_top
...dont_touch_spare
...ont_touch_out_err
...ont_touch_in_start
...dont_touch_in_clk
...dont_touch_in_rst
...t_touch_vssIO_top
...ont_touch_dout23
...ont_touch_dout22
...ont_touch_dout21
...ont_touch_dout20
...ont_touch_dout19
...ont_touch_dout18
...uch_corner_ur
...dont_touch_din17
...dont_touch_din16
...dont_touch_din15
...dont_touch_din14
...dont_touch_din13
...dont_touch_din12
...ouch_vssCore_left
...dont_touch_din11
...dont_touch_din10
...dont_touch_din09
...dont_touch_din08
...dont_touch_din07
...dont_touch_din06
...ouch_vddCore_left
...dont_touch_din05
...dont_touch_din04
...dont_touch_din03
...dont_touch_din02
...dont_touch_din01
...dont_touch_din00
...ont_touch_dout17
...ont_touch_dout16
...ont_touch_dout15
...ont_touch_dout14
...ont_touch_dout13
...ont_touch_dout12
...uch_vssCore_right
...ont_touch_dout11
...ont_touch_dout10
...ont_touch_dout09
...ont_touch_dout08
...ont_touch_dout07
...ont_touch_dout06
...uch_vddCore_right
...ont_touch_dout05
...ont_touch_dout04
...ont_touch_dout03
...ont_touch_dout02
...ont_touch_dout01
...ont_touch_dout00
...uch_corner_ll
...dont_touch_din24
...dont_touch_din25
...dont_touch_din26
...dont_touch_din27
...dont_touch_din28
...dont_touch_din29
...uch_vddIO_bottom
...dont_touch_din30
...dont_touch_din31
...dont_touch_done
...ont_touch_dout31
...ont_touch_dout30
...uch_vssIO_bottom
...ont_touch_dout29
...ont_touch_dout28
...ont_touch_dout27
...ont_touch_dout26
...ont_touch_dout25
...ont_touch_dout24
...uch_corner_lr

Place



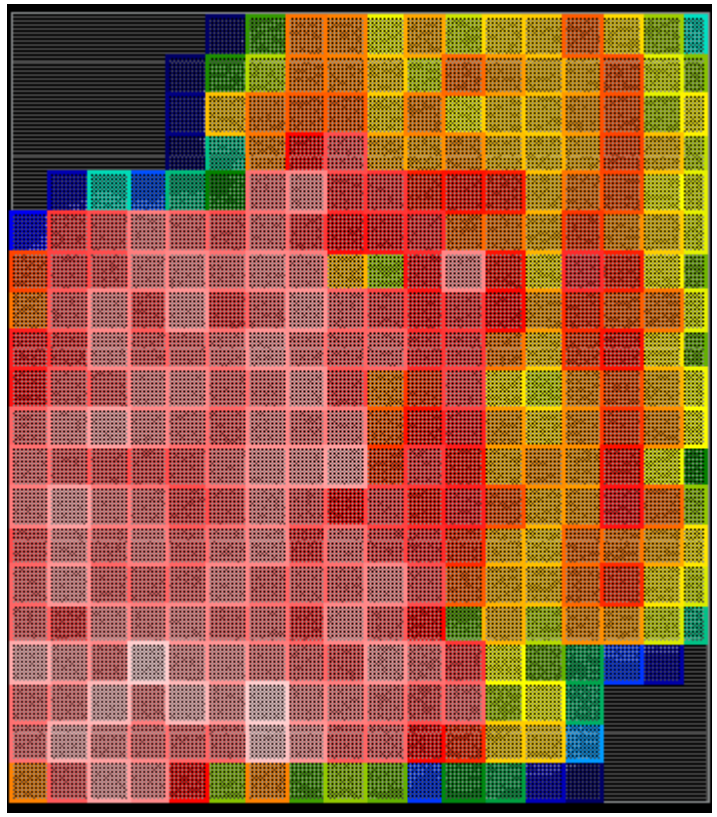
Place clock buf root center. Because Clock buf root will minimize distance from clock root to sink

place



ALU

Ram_wrapper_cell



Densitymap

Time_design –pre_cts

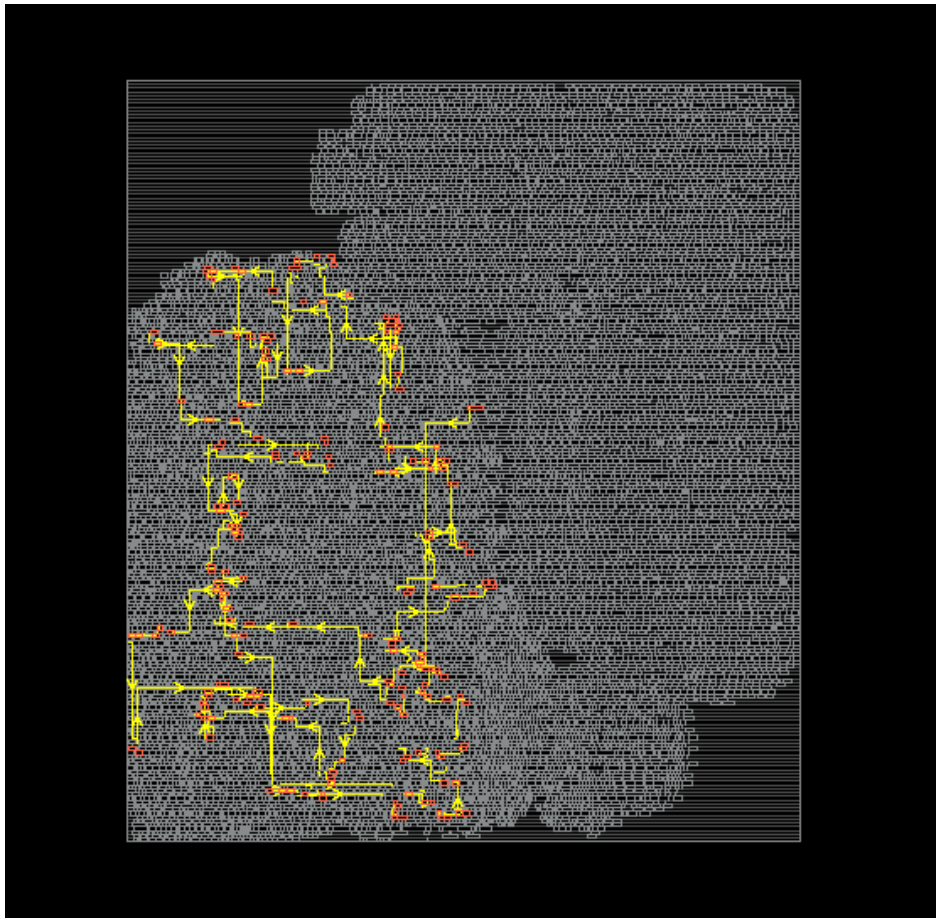
```
------------------------------------------------------------
        time_design Summary
------------------------------------------------------------

Setup views included:
 default_emulate_view

+--------------------+--------+---------+---------+---------+---------+---------+---------+---------+
|     Setup mode     |   all  | reg2reg | in2out  | default | in2icg  |  input  | output  | reg2icg |
+--------------------+--------+---------+---------+---------+---------+---------+---------+---------+
|          WNS (ns):| -0.110 | -0.110  |   N/A   |  0.000  |  0.626  |  0.248  |  0.459  |  0.509  |
|          TNS (ns):| -0.200 | -0.200  |   N/A   |  0.000  |  0.000  |  0.000  |  0.000  |  0.000  |
|   Violating Paths:|   2    |    2    |   N/A   |    0    |    0    |    0    |    0    |    0    |
|         All Paths:|  4525  |  4356   |   N/A   |    0    |    1    |   32    |   35    |   134   |
+--------------------+--------+---------+---------+---------+---------+---------+---------+---------+

+----------------+-----------------------------+-----------------+
|                |              Real           |       Total     |
|    DRVs        +-----------------+-----------+-----------------|
|                |  Nr nets(terms) | Worst Vio |  Nr nets(terms) |
+----------------+-----------------+-----------+-----------------+
|   max_cap      |      0 (0)      |   0.000   |      0 (0)      |
|   max_tran     |      0 (0)      |   0.000   |      0 (0)      |
|   max_fanout   |     70 (70)     |    -14    |     71 (71)     |
|   max_length   |      0 (0)      |     0     |      0 (0)      |
+----------------+-----------------+-----------+-----------------+

Density: 72.900%
Routing Overflow: 0.00% H and 0.00% V
```

```
------------------------------------------------------------
        time_design Summary
------------------------------------------------------------

Hold views included:
 default_emulate_view

+--------------------+--------+---------+---------+---------+---------+---------+---------+---------+
|     Hold mode      |   all  | reg2reg | in2out  | default | in2icg  |  input  | output  | reg2icg |
+--------------------+--------+---------+---------+---------+---------+---------+---------+---------+
|          WNS (ns):| -0.079 | -0.079  |   N/A   |  0.000  |  1.276  |  1.053  |  1.280  |  0.338  |
|          TNS (ns):| -0.116 | -0.116  |   N/A   |  0.000  |  0.000  |  0.000  |  0.000  |  0.000  |
|   Violating Paths:|   2    |    2    |   N/A   |    0    |    0    |    0    |    0    |    0    |
|         All Paths:|  4525  |  4356   |   N/A   |    0    |    1    |   32    |   35    |   134   |
+--------------------+--------+---------+---------+---------+---------+---------+---------+---------+

Density: 72.900%
Routing Overflow: 0.00% H and 0.00% V
------------------------------------------------------------
```

Setup Timing: The WNS has significantly improved to -0.110ns with only 2 violating paths remaining in the reg2reg group.

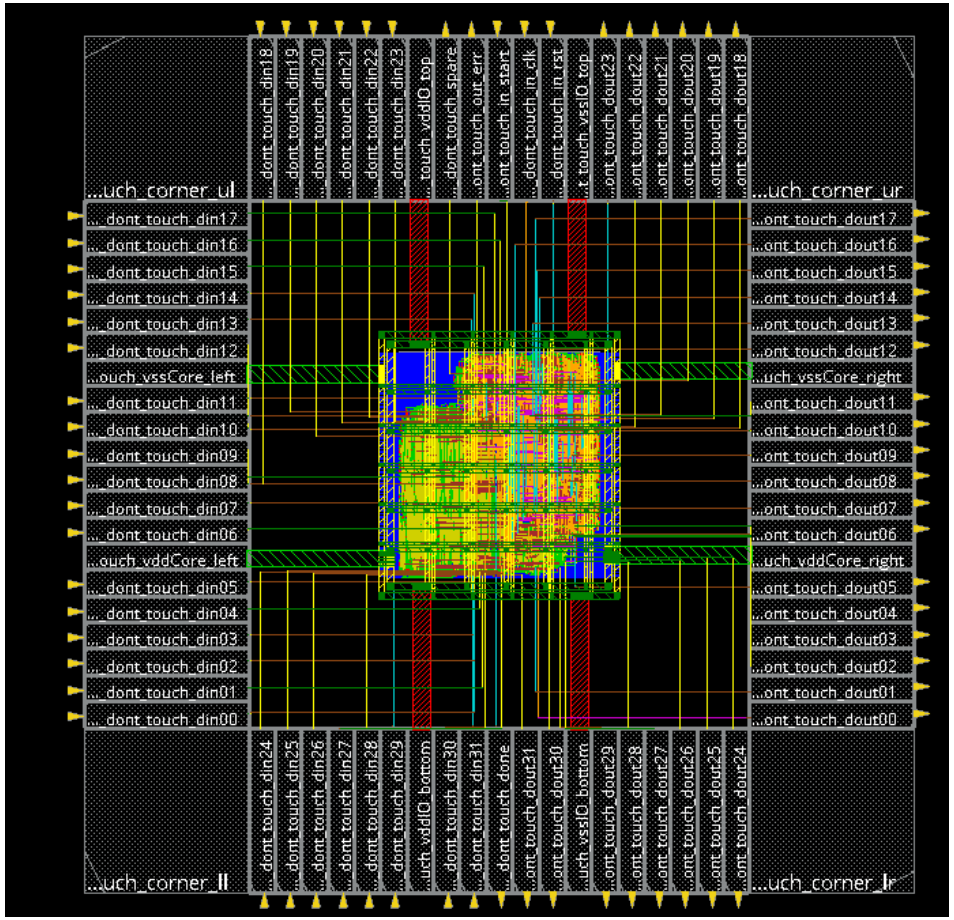
Hold Timing: The design now shows a slight violation with a WNS of -0.079ns and 2 violating paths

Worst datapath



```
Path 1: VIOLATED (-0.110 ns) Setup Check with Pin U_CORE/DP/ACC_reg/qout_regx0x/CK->D
               View: default_emulate_view
              Group: reg2reg
         Startpoint: (R) U_CORE/DP/MDR_reg/qout_regx7x/CK
              Clock: (R) clk
           Endpoint: (R) U_CORE/DP/ACC_reg/qout_regx0x/D
              Clock: (R) clk

                       Capture       Launch
        Clock Edge:+    12.000        0.000
       Src Latency:+     0.000        0.000
       Net Latency:+     0.000 (I)   -0.300 (I)
           Arrival:=    12.000       -0.300

             Setup:-     0.120
     Required Time:=    11.880
      Launch Clock:=    -0.300
         Data Path:+    12.289
             Slack:=    -0.110
```

CTS


...uch_corner_ul
...uch_corner_ur
...uch_corner_ll
...uch_corner_lr
...dont touch din18
...dont touch din19
...dont touch din20
...dont touch din21
...dont touch din22
...dont touch din23
...touch vddIO top
...dont touch spare
...ont touch out err
...ont touch in start
...dont touch in clk
...dont touch in rst
...t touch vssIO top
...ont touch dout23
...ont touch dout22
...ont touch dout21
...ont touch dout20
...ont touch dout19
...ont touch dout18
...dont touch din17
...dont touch din16
...dont touch din15
...dont touch din14
...dont touch din13
...dont touch din12
...uch vssCore left
...dont touch din11
...dont touch din10
...dont touch din09
...dont touch din08
...dont touch din07
...dont touch din06
...uch vddCore left
...dont touch din05
...dont touch din04
...dont touch din03
...dont touch din02
...dont touch din01
...dont touch din00
...ont touch dout17
...ont touch dout16
...ont touch dout15
...ont touch dout14
...ont touch dout13
...ont touch dout12
...uch vssCore right
...ont touch dout11
...ont touch dout10
...ont touch dout09
...ont touch dout08
...ont touch dout07
...ont touch dout06
...uch vddCore right
...ont touch dout05
...ont touch dout04
...ont touch dout03
...ont touch dout02
...ont touch dout01
...ont touch dout00
...dont touch din24
...dont touch din25
...dont touch din26
...dont touch din27
...dont touch din28
...dont touch din29
...uch vddIO bottom
...dont touch din30
...dont touch din31
...dont touch done
...ont touch dout31
...ont touch dout30
...uch vssIO bottom
...ont touch dout29
...ont touch dout28
...ont touch dout27
...ont touch dout26
...ont touch dout25
...ont touch dout24

Report_skew_group

Skew Group Structure:
==========

Skew Group	Sources	Constrained Sinks	Unconstrained Sinks
clk/default_emulate_constraint_mode	1	4263	0

Skew Group Summary:
==========

Timing Corner	Skew Group	ID Target	Min ID	Max ID	Avg ID	Std.Dev. ID	Skew Target Type	Skew Target	Skew	Skew window occupancy
default_emulate_delay_corner:both.early	clk/default_emulate_constraint_mode	-	0.686	0.920	0.726	0.021	ignored	-	0.234	-
default_emulate_delay_corner:both.late	clk/default_emulate_constraint_mode	none	0.687	0.940	0.728	0.022	explicit	*0.030	0.252	85.1% {0.707, 0.737}
default_emulate_power_delay_corner:setup.early	clk/default_emulate_constraint_mode	-	0.686	0.920	0.726	0.021	ignored	-	0.234	-
default_emulate_power_delay_corner:setup.late	clk/default_emulate_constraint_mode	-	0.687	0.922	0.727	0.021	ignored	-	0.235	-

Skew target 0.03 with 85.1% is good

Time_design –post_cts

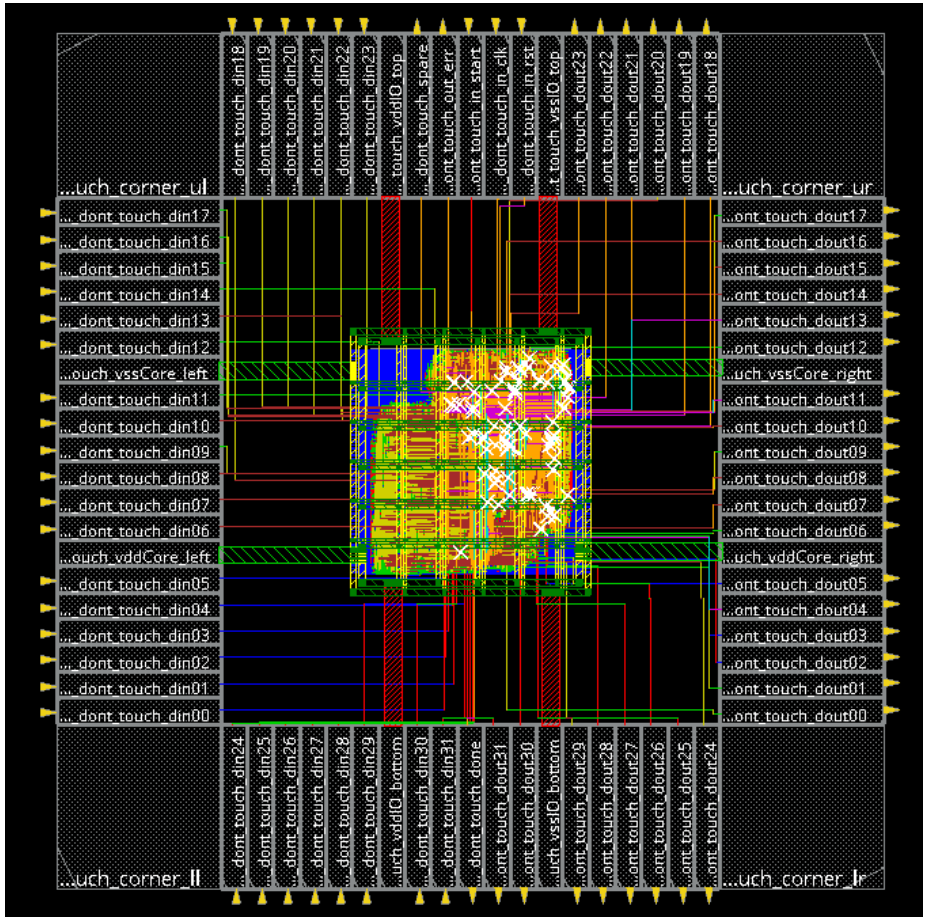
```
------------------------------------------------------------
         time_design Summary
------------------------------------------------------------

Setup views included:
 default_emulate_view

+--------------------+--------------------------------------------------+
|     ID Number      |                  Path Group Name                 |
+--------------------+--------------------------------------------------+
|         1          |reg2ckgate                                        |
+--------------------+--------------------------------------------------+

+--------------------+---------+---------+---------+---------+---------+---------+---------+
|     Setup mode     |   all   | reg2reg | in2reg  | reg2out | in2out  | default |1:reg..te|
+--------------------+---------+---------+---------+---------+---------+---------+---------+
|           WNS (ns):| -0.050  | -0.050  |  0.302  |  0.063  |   N/A   |  0.415  |  0.027  |
|           TNS (ns):| -0.207  | -0.207  |  0.000  |  0.000  |   N/A   |  0.000  |  0.000  |
|    Violating Paths:|   12    |   12    |    0    |    0    |   N/A   |    0    |    0    |
|          All Paths:|  4525   |  4265   |   33    |   35    |   N/A   |   91    |   134   |
+--------------------+---------+---------+---------+---------+---------+---------+---------+

+----------------+-----------------------------+-----------------+
|                |             Real            |       Total     |
|    DRVs        +-----------------+-----------+-----------------|
|                |  Nr nets(terms) | Worst Vio |  Nr nets(terms) |
+----------------+-----------------+-----------+-----------------+
|   max_cap      |      0 (0)      |   0.000   |      0 (0)      |
|   max_tran     |      0 (0)      |   0.000   |      0 (0)      |
|   max_fanout   |      0 (0)      |     0     |      0 (0)      |
|   max_length   |      0 (0)      |     0     |      0 (0)      |
+----------------+-----------------+-----------+-----------------+

Density: 69.470%
Routing Overflow: 0.00% H and 0.00% V
```

```
------------------------------------------------------------
         time_design Summary
------------------------------------------------------------

Hold views included:
 default_emulate_view

+--------------------+--------------------------------------------------+
|     ID Number      |                  Path Group Name                 |
+--------------------+--------------------------------------------------+
|         1          |reg2ckgate                                        |
+--------------------+--------------------------------------------------+

+--------------------+---------+---------+---------+---------+---------+---------+---------+
|     Hold mode      |   all   | reg2reg | in2reg  | reg2out | in2out  | default |1:reg..te|
+--------------------+---------+---------+---------+---------+---------+---------+---------+
|           WNS (ns):| -0.009  | -0.009  |  0.924  |  1.415  |   N/A   |  0.856  |  0.689  |
|           TNS (ns):| -0.009  | -0.009  |  0.000  |  0.000  |   N/A   |  0.000  |  0.000  |
|    Violating Paths:|    1    |    1    |    0    |    0    |   N/A   |    0    |    0    |
|          All Paths:|  4525   |  4265   |   33    |   35    |   N/A   |   91    |   134   |
+--------------------+---------+---------+---------+---------+---------+---------+---------+

Density: 69.470%
Routing Overflow: 0.00% H and 0.00% V
```

Setup Timing: The design has a small WNS of -0.050ns with 12 violating paths, all concentrated within the reg2reg group.

Hold Timing: There is a near-zero violation with a WNS of -0.009ns on only 1 violating path

Route


...uch_corner_ul
...dont touch din17
...dont touch din16
...dont touch din15
...dont touch din14
...dont touch din13
...dont touch din12
...ouch vssCore left
...dont touch din11
...dont touch din10
...dont touch din09
...dont touch din08
...dont touch din07
...dont touch din06
...ouch vddCore left
...dont touch din05
...dont touch din04
...dont touch din03
...dont touch din02
...dont touch din01
...dont touch din00
...uch_corner_ll
...dont touch din24
...dont touch din25
...dont touch din26
...dont touch din27
...dont touch din28
...dont touch din29
...uch vddIO bottom
...dont touch din30
...dont touch din31
...dont touch done
...ont touch dout31
...ont touch dout30
...uch vssIO bottom
...ont touch dout29
...ont touch dout28
...ont touch dout27
...ont touch dout26
...ont touch dout25
...ont touch dout24
...uch_corner_lr
...ont touch dout17
...ont touch dout16
...ont touch dout15
...ont touch dout14
...ont touch dout13
...ont touch dout12
...uch vssCore right
...ont touch dout11
...ont touch dout10
...ont touch dout09
...ont touch dout08
...ont touch dout07
...ont touch dout06
...uch vddCore right
...ont touch dout05
...ont touch dout04
...ont touch dout03
...ont touch dout02
...ont touch dout01
...ont touch dout00
...uch_corner_ur
...dont touch din18
...dont touch din19
...dont touch din20
...dont touch din21
...dont touch din22
...dont touch din23
...touch vddIO top
...dont touch spare
...ont touch out err
...ont touch in start
...dont touch in clk
...dont touch in rst
...t touch vssIO top
...ont touch dout23
...ont touch dout22
...ont touch dout21
...ont touch dout20
...ont touch dout19
...ont touch dout18

Check_drc -limit 0 & Check_connectivity

```
 Verification Complete : 3443 Viols.

 Violation Summary By Layer and Type:

                  Short   MetSpc      Mar   EOLSpc   Totals
        Metal1      249       37        1        0      287
        Metal2      907       72        1        0      980
        Metal3     1747       72        1        2     1822
        Metal4      228       45        0        0      273
        Metal5       52       18        0        0       70
        Metal6        8        2        0        0       10
        Metal7        1        0        0        0        1
        Totals     3192      246        3        2     3443

 *** End Verify DRC (CPU TIME: 0:01:27  ELAPSED TIME: 0:01:31  MEM: 291.1M) ***
```

```
Begin Summary
    838 Problem(s) (IMPVFC-98): Net has no global routing and no special routing.
    162 Problem(s) (IMPVFC-96): Terminal(s) are not connected.
    1000 total info(s) created.
End Summary

End Time: Thu Apr 16 11:16:46 2026
Time Elapsed: 0:00:00.0

******** End: VERIFY CONNECTIVITY ********
  Verification Complete : 1000 Viols.  0 Wrngs.
  (CPU Time: 0:00:00.1  MEM: 0.000M)
```

3192 short
246 spacing
3 Mar
2 EOLspc
More than 1000 vio connectivity
=> route_eco and source fix_drc.tcl

Check timming

```
-----------------------------------------------------------------
         time_design Summary
-----------------------------------------------------------------

Setup views included:
 default_emulate_view

+--------------------+-------------------------------------------------+
|     ID Number      |                Path Group Name                  |
+--------------------+-------------------------------------------------+
|         1          |reg2ckgate                                       |
+--------------------+-------------------------------------------------+

+--------------------+---------+---------+---------+---------+---------+---------+---------+
|     Setup mode     |   all   | reg2reg | in2reg  | reg2out | in2out  | default |1:reg..te|
+--------------------+---------+---------+---------+---------+---------+---------+---------+
|           WNS (ns):| -0.792  | -0.792  |  0.302  | -0.003  |   N/A   |  0.376  | -0.012  |
|           TNS (ns):| -15.177 | -15.162 |  0.000  | -0.003  |   N/A   |  0.000  | -0.012  |
|    Violating Paths:|   102   |   100   |    0    |    1    |   N/A   |    0    |    1    |
|          All Paths:|  4525   |  4265   |   33    |   35    |   N/A   |   91    |   134   |
+--------------------+---------+---------+---------+---------+---------+---------+---------+

+----------------+-----------------------------+------------------+
|                |             Real            |       Total      |
|    DRVs        +------------------+----------+------------------|
|                |  Nr nets(terms)  | Worst Vio |  Nr nets(terms)  |
+----------------+------------------+----------+------------------+
|   max_cap      |      0 (0)       |  0.000   |      0 (0)       |
|   max_tran     |      2 (27)      |  -0.013  |      2 (27)      |
|   max_fanout   |      0 (0)       |    0     |      0 (0)       |
|   max_length   |      0 (0)       |    0     |      0 (0)       |
+----------------+------------------+----------+------------------+

Density: 69.470%
-----------------------------------------------------------------
```

```
-----------------------------------------------------------------
         time_design Summary
-----------------------------------------------------------------

Hold views included:
 default_emulate_view

+--------------------+-------------------------------------------------+
|     ID Number      |                Path Group Name                  |
+--------------------+-------------------------------------------------+
|         1          |reg2ckgate                                       |
+--------------------+-------------------------------------------------+

+--------------------+---------+---------+---------+---------+---------+---------+---------+
|     Hold mode      |   all   | reg2reg | in2reg  | reg2out | in2out  | default |1:reg..te|
+--------------------+---------+---------+---------+---------+---------+---------+---------+
|           WNS (ns):|  0.005  |  0.005  |  0.955  |  1.379  |   N/A   |  0.889  |  0.734  |
|           TNS (ns):|  0.000  |  0.000  |  0.000  |  0.000  |   N/A   |  0.000  |  0.000  |
|    Violating Paths:|    0    |    0    |    0    |    0    |   N/A   |    0    |    0    |
|          All Paths:|  4525   |  4265   |   33    |   35    |   N/A   |   91    |   134   |
+--------------------+---------+---------+---------+---------+---------+---------+---------+

Density: 69.470%
-----------------------------------------------------------------
```

- **Setup Timing:** The design is failing with a significant **WNS of -0.792ns** and **102 violating paths**, primarily within the reg2reg group.
- **Hold Timing:** The design is clean with a positive **WNS of 0.005ns** and **zero violations**, indicating no hold-time risks at this stage.

=> route_eco & opt_design

Route_eco and source fix_drc.tcl

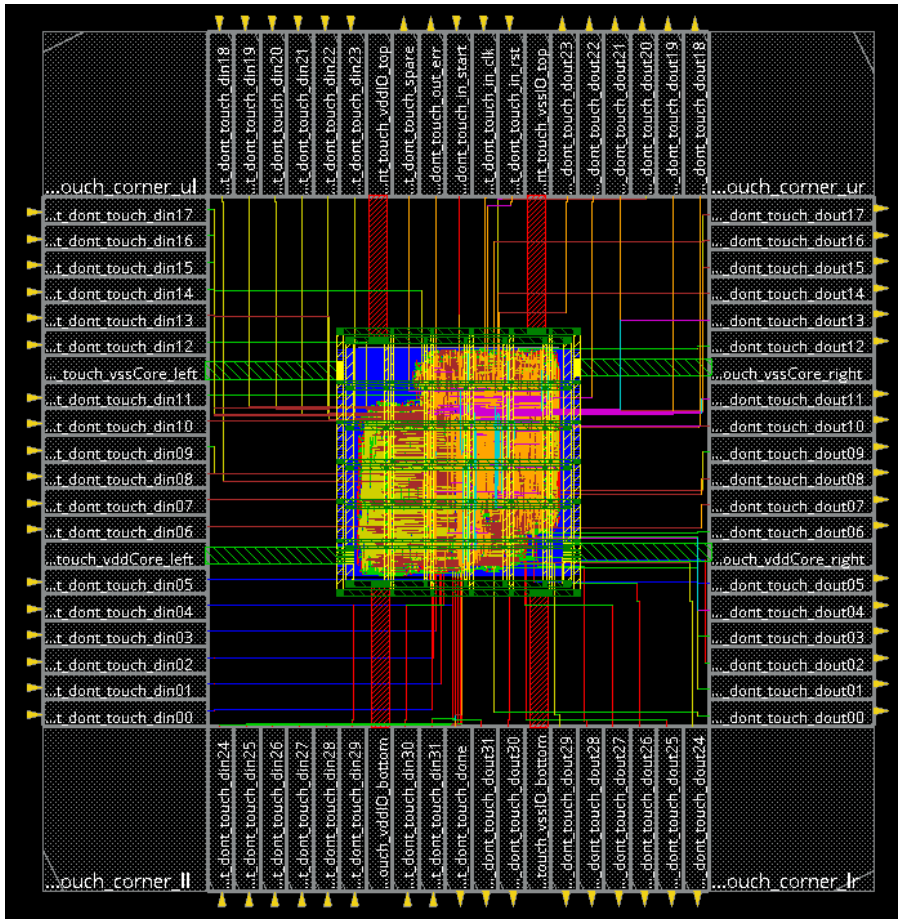
```
 Verification Complete : 49 Viols.

Violation Summary By Layer and Type:

                 Short   MetSpc   Totals
       Metal1       44        5       49
       Totals       44        5       49

*** End Verify DRC (CPU TIME: 0:01:29  ELAPSED TIME: 0:01:34  MEM: 256.1M) ***
```

```
******** End: VERIFY CONNECTIVITY ********
 Verification Complete : 0 Viols.  0 Wrngs.
 (CPU Time: 0:00:01.1  MEM: 0.000M)
```

- 44 short
- 5 spacing
- 0 connectivity

=> Can fix by manual

Chip finish




```
******** End: VERIFY CONNECTIVITY ********
  Verification Complete : 0 Viols.  0 Wrngs.
  (CPU Time: 0:00:01.0  MEM: 0.000M)


  Verification Complete : 0 Viols.

 *** End Verify DRC (CPU TIME: 0:00:02.2  ELAPSED TIME: 0:00:04.0  MEM: 272.1M) ***
```

Check_timming

ID Number	Path Group Name
1	reg2ckgate

Setup mode	all	reg2reg	in2reg	reg2out	in2out	default	1:reg..te
WNS (ns):	-0.008	-0.008	0.313	0.113	N/A	0.445	0.113
TNS (ns):	-0.021	-0.021	0.000	0.000	N/A	0.000	0.000
Violating Paths:	3	3	0	0	N/A	0	0
All Paths:	4525	4265	33	35	N/A	91	134

DRVs	Real		Total
	Nr nets(terms)	Worst Vio	Nr nets(terms)
max_cap	0 (0)	0.000	0 (0)
max_tran	0 (0)	0.000	0 (0)
max_fanout	0 (0)	0	0 (0)
max_length	0 (0)	0	0 (0)

Density: 69.477%

ID Number	Path Group Name
1	reg2ckgate

Hold mode	all	reg2reg	in2reg	reg2out	in2out	default	1:reg..te
WNS (ns):	-0.000	-0.000	0.943	1.364	N/A	0.781	0.709
TNS (ns):	-0.000	0.000	0.000	0.000	N/A	0.000	0.000
Violating Paths:	1	1	0	0	N/A	0	0
All Paths:	4525	4265	33	35	N/A	91	134

Density: 69.477%

- **Hold Timing:** Timing is practically met with a negligible **WNS of -0.000ns** on **1 violating path**
- **Setup Timing:** The design is nearly closed with a marginal **WNS of -0.008ns** and only **3 violating paths**